MOLECULAR DYNAMICS REPORT

# Effect of pH and Salt on the Aggregation of Silica Nanoparticles in Water

CLAYFF + SPC water | LAMMPS | two 3 nm alpha-quartz nanoparticles

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Two 3 nm alpha-quartz silica nanoparticles in explicit SPC water, simulated with the CLAYFF force field in LAMMPS. The study varies surface charge (pH 2 to 12) and adds NaCl or $CaCl_2$ at pH 10, and measures how the separation of the two particles evolves.

## 1. Summary

Two crystalline silica nanoparticles (3 nm diameter, carved from alpha-quartz and capped with silanol groups) were placed 4 nm apart in water. Surface charge was set from pH with a Henderson–Hasselbalch deprotonation model (pKa 6.8), and $Na^+$ counterions were added for neutrality. Each condition was run once for 35–55 ps and the particle–particle centre-of-mass distance was tracked.

- **Neutral surface (pH 2, pH 4):** the separation oscillates around 40.4 Å with no net drift.
- **Partly charged (pH 7, 61% deprotonated):** the separation drifts outward by about +0.36 Å over 50 ps, the clearest sign of repulsion in the data.
- **Highly charged (pH 8, 10, 12):** pH 8 drifts outward by roughly +0.4 Å. pH 10 and pH 12 show only about +0.08 Å and fluctuate within a 0.3 Å band. $Na^+$ accumulates more strongly near the pH 10 surface than near the pH 7 surface (RDF peak 0.038 vs 0.020), which is a plausible explanation but not a proof.
- **Salt (NaCl, $CaCl_2$ at 0.05 and 0.3 M, pH 10):** the four runs stay within 40.3–41.0 Å and show no resolvable trend with concentration or ion valence in these short runs.

These are single, short, qualitative trajectories. The drifts are comparable to the thermal fluctuation of each trace, so the results indicate trends to be tested with longer runs and replicas, not measured effects. Section 7 lists the limitations in full.

## 2. Motivation and relevance to ceramic processing

Whether fine oxide particles stay dispersed or aggregate in water controls slip viscosity, green-body packing and, after firing, microstructure. In water the surface charge of silica comes from silanol groups ($Si–OH \rightleftarrows Si–O^- + H^+$), so pH sets the charge, and dissolved ions screen it (DLVO theory, Schulze–Hardy rule).

This is directly relevant to **refractory castables**. Colloidal silica (silica sol) and microsilica are used as binders and fines in low-cement and cement-free castables, and gelation or deflocculation of the silica is governed by pH and by $Ca^{2+}$ released from calcium aluminate cement. Castable pore water sits at pH

11–12 and is rich in $Ca^{2+}$, which is the regime that the pH 10/12 and $CaCl_2$ runs here are meant to represent. This work is a molecular-scale model of that mechanism at room temperature in water. It does not address high-temperature refractory properties such as thermal shock, creep or slag attack.

# 3. Model and methods

## 3.1 Nanoparticle construction

An alpha-quartz CIF was read with ASE, replicated into a supercell and cut to a 3 nm sphere. Si–O coordination was found with a k-d tree (2.0 Å cutoff). Debris was removed, and under-coordinated Si atoms were completed to tetrahedral geometry with new oxygens. Every terminal oxygen was capped with a hydrogen to form silanol (Si–OH). A final pass nudged any over-coordinated oxygens outward. The result is 1497 atoms per particle ($Si_{382}O_{881}H_{234}$), with 234 surface silanol groups on a surface area of about 28 $nm^2$.

## 3.2 pH-dependent surface charge

The fraction of deprotonated silanols is 1/(1 + 10^(pKa − pH)) with pKa = 6.8. Deprotonated sites lose their hydrogen, take the CLAYFF charge −1.525 e, and are paired with one $Na^+$ each in the solvent. The sites are chosen at random (fixed seed).

| pH | Deprotonated sites / NP | Net charge / NP (e) | Comment |
|---|---|---|---|
| 2 | 0 of 234 (0%) | 0 | Neutral surface |
| 4 | 0 of 234 (0.2% → 0 sites) | 0 | Same system as pH 2 after rounding |
| 7 | 143 of 234 (61.3%) | −143 | Partly charged |
| 8 | 220 of 234 (94.1%) | −220 | Nearly fully charged |
| 10 | 234 of 234 (99.9%) | −234 | Fully charged |
| 12 | 234 of 234 (100%) | −234 | Same system as pH 10 (model saturates) |

Because the model rounds to whole sites, pH 4 is identical to pH 2 and pH 12 is identical to pH 10. The pH sweep therefore contains four distinct charge states (0, 143, 220 and 234 charges per particle). The pH 4 and pH 12 runs reproduced the pH 2 and pH 10 traces and are best read as repeat runs of the same system.

## 3.3 System assembly

Two identical particles were placed 40 Å apart (centre to centre) along x in a periodic box with 6 Å padding. For the pH 10 system the box is 82.8 × 42.3 × 41.3 Å (14,634 atoms). Packmol filled the free volume with SPC water at 0.0334 molecules/$Å^3$ and placed the counterions, excluding a 16 Å sphere around each particle centre. Atom counts range from 14,634 (pH 10/12) to 15,990 (pH 2).

## 3.4 Force field

CLAYFF nonbonded Si–O framework with harmonic O–H bonds on silanols, and rigid SPC water (SHAKE). Lorentz–Berthelot (arithmetic) mixing, 10 Å cutoff, PPPM long-range electrostatics ($10^{-4}$ relative accuracy).

| Type | Species | ε (kcal/mol), σ (Å) | Charge (e) |
|---|---|---|---|
| 1 | Si | 1.8405e-6, 3.7064 | +2.1 |
| 2 / 3 / 8 | O bridging / silanol / deprotonated | 0.1554, 3.5532 | −1.05 / −0.95 / −1.525 |
| 4 | H (silanol) | 0, 0 | +0.425 |
| 5 / 6 | O, H (SPC water) | 0.1554, 3.166 / 0, 0 | −0.82 / +0.41 |
| 7 | $Na^+$ | 0.1301, 2.6378 | +1 |
| 9 / 10 | $Cl^-$ / $Ca^{2+}$ (salt study) | 0.1000, 4.400 / 0.1000, 2.872 | −1 / +2 |

Bond: K = 554.13 kcal/mol/Å², $r_0$ = 1.0 Å. Angle: K = 45.77 kcal/mol/rad², $\theta_0$ = 109.47°. The $Cl^-$ and $Ca^{2+}$ parameters are borrowed from other ion models and were not tuned to CLAYFF.

### 3.5 Simulation protocol

- Conjugate-gradient minimization, then velocities at 300 K.
- 20 ps Langevin equilibration with the thermostat ramped from 10 to 300 K (NVE + Langevin).
- NVT production at 300 K (Nosé–Hoover), 1 fs timestep, 35–55 ps depending on the run.
- The distance between the two particles' centres of mass was written every 100 steps; atomic positions were dumped every 1000 steps.

Runs used 2 OpenMP threads on a free Google Colab CPU runtime.

## 4. Results: pH sweep

Each plot shows the centre-to-centre distance during production. The starting separation is 40 Å and the surface-to-surface gap is about 10 Å. Fluctuations of 0.2–0.4 Å are present in every run, so a drift only means something if it is larger than that.

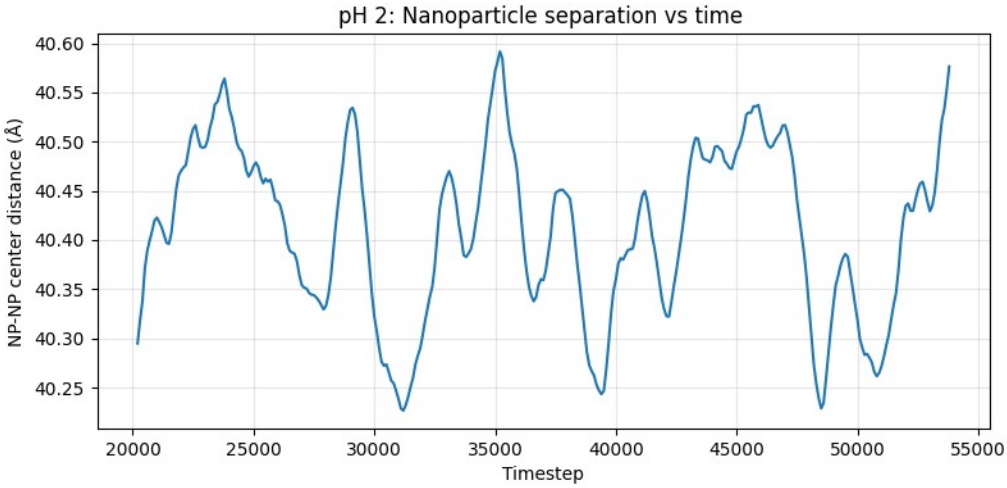



*Figure 1. pH 2 (neutral surface): the separation oscillates between about 40.2 and 40.6 Å with no net drift.*

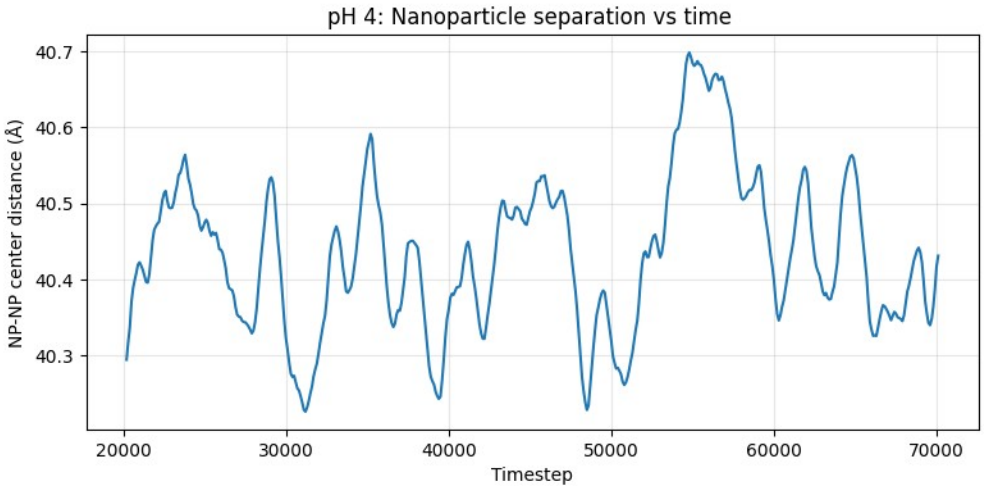



*Figure 2. pH 4: same charge state as pH 2 and the same behaviour (no net drift).*

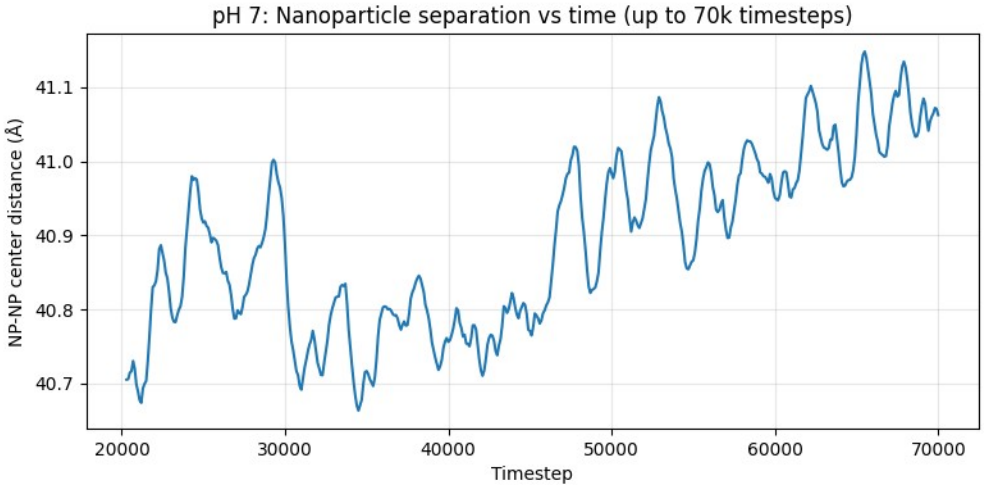



*Figure 3. pH 7 (61% deprotonated): the separation rises from about 40.7 to 41.1 Å, a net drift of roughly +0.36 Å.*

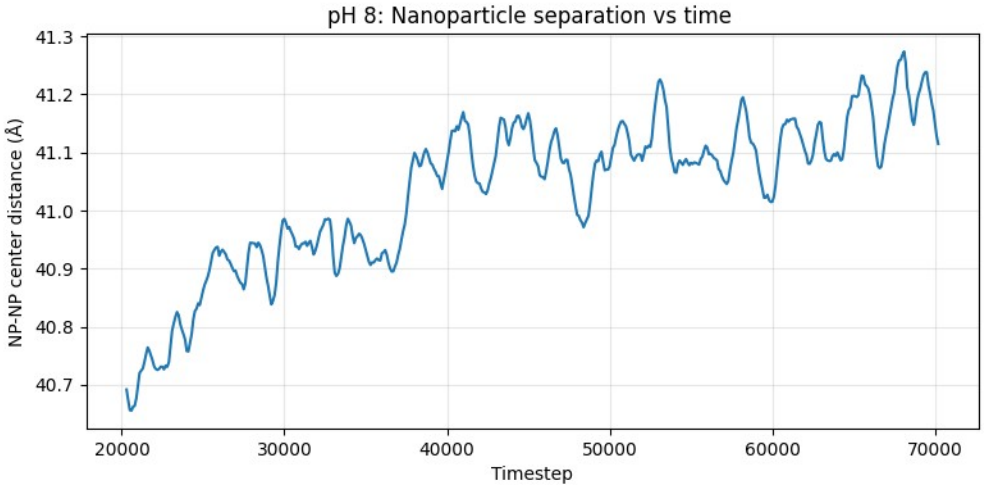



*Figure 4. pH 8 (94% deprotonated): the separation rises from about 40.7 to 41.1–41.2 Å, a net drift of roughly +0.4 Å, then levels off.*

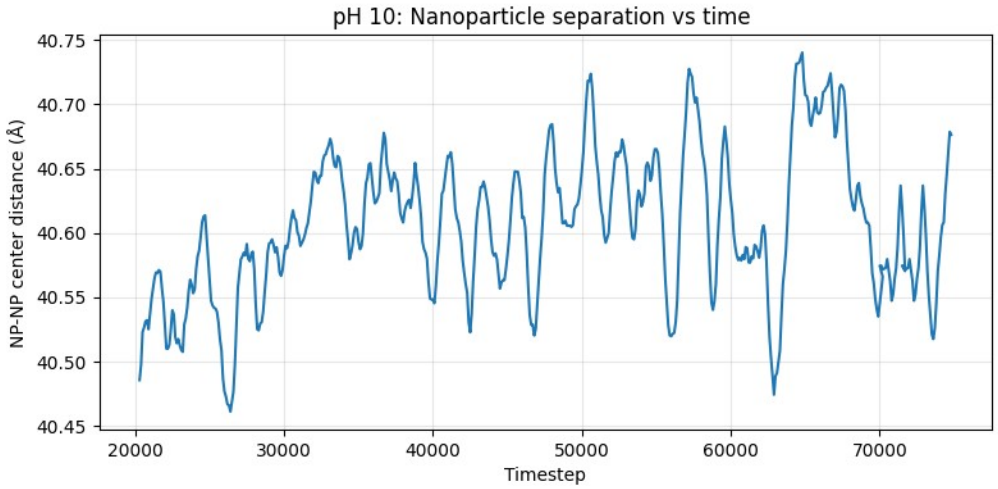



*Figure 5. pH 10 (fully deprotonated): only a small rise (about +0.08 Å between block averages) and fluctuations of 40.46–40.74 Å.*

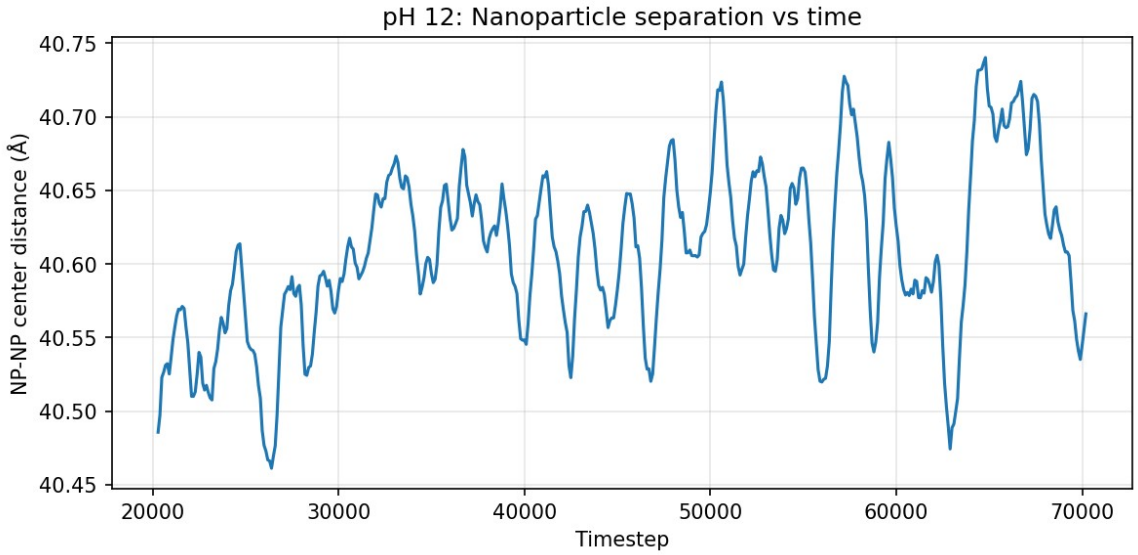



*Figure 6. pH 12: the same charge state as pH 10, and the trace reproduces it.*

## 4.1 Reading the pH sweep

| Condition | Net drift | Observation |
|---|---|---|
| pH 2 / pH 4 | ≈ 0 | Oscillation only; no sustained change in separation |
| pH 7 | ≈ +0.36 Å | Steady outward drift over 50 ps |
| pH 8 | ≈ +0.4 Å | Outward drift in the first 40 ps, then a plateau |
| pH 10 / pH 12 | ≈ +0.08 Å | Small drift, within the thermal fluctuation |

The neutral case is flat and the partly charged cases move outward, which is the direction expected if charge produces repulsion. The fully charged cases do not move further than pH 7. The pH 8 value is a drift read from a single trace and cannot be separated from the pH 7 and pH 10 results. A distinct “peak” at pH 8 should not be claimed.

## 5. Why is pH 10 not more repulsive than pH 7?

One explanation is counterion condensation. At full charge each particle carries 234 negative sites in a small box, so many $Na^+$ ions sit within a few Ångströms of the surface and partly neutralise it. The net force on the other particle is then smaller than the bare charge suggests (charge renormalisation, as in the Poisson–Boltzmann cell model). A radial distribution of $Na^+$ around surface oxygens was computed from the last 20 frames of the pH 7 and pH 10 trajectories.

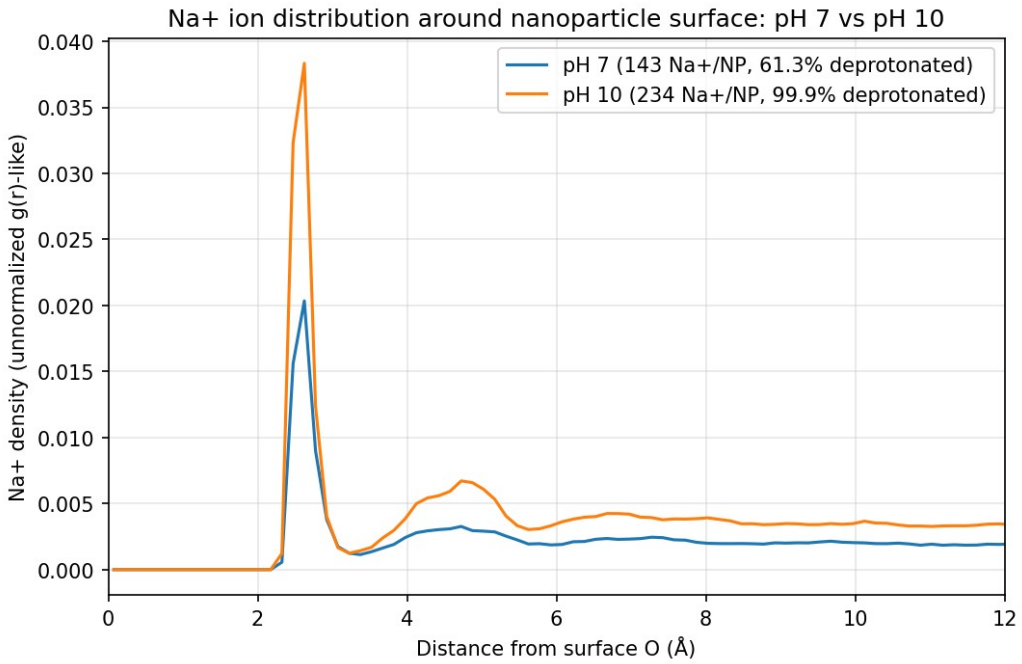



*Figure 7. $Na^+$ density versus distance from surface oxygens. The first peak is 0.038 at pH 10 and 0.020 at pH 7 (unnormalised, arbitrary units).*

The taller first peak at pH 10 agrees with stronger $Na^+$ binding at the more highly charged surface. It does not establish the mechanism: the pH 10 particle has more charge and more $Na^+$, so a taller peak is expected on any account, and the histogram is not normalised to the bulk density. The drift difference (+0.36 vs +0.08 Å) is also of similar size to the run-to-run fluctuation. The explanation is a hypothesis consistent with the data.

## 6. Results: salt study at pH 10

Using the pH 10 particles and box, background salt was added on top of the 468 neutralising $Na^+$ ions. NaCl and $CaCl_2$ were each run at 0.05 M and 0.3 M. In the 116,000 $Å^3$ of free volume these correspond to only 3 and 21 formula units, so the ion statistics are poor. Each run is 10 ps equilibration followed by 25 ps production.

| System | Atoms | Ions added | Separation range (Å, from plot) |
|---|---|---|---|
| NaCl 0.05 M | 14,640 | 3 $Na^+$, 3 $Cl^-$ | 40.5 – 40.9 |
| NaCl 0.3 M | 14,676 | 21 $Na^+$, 21 $Cl^-$ | 40.6 – 41.0 |
| $CaCl_2$ 0.05 M | 14,643 | 3 $Ca^{2+}$, 6 $Cl^-$ | 40.3 – 40.6 |
| $CaCl_2$ 0.3 M | 14,697 | 21 $Ca^{2+}$, 42 $Cl^-$ | 40.6 – 40.8 |

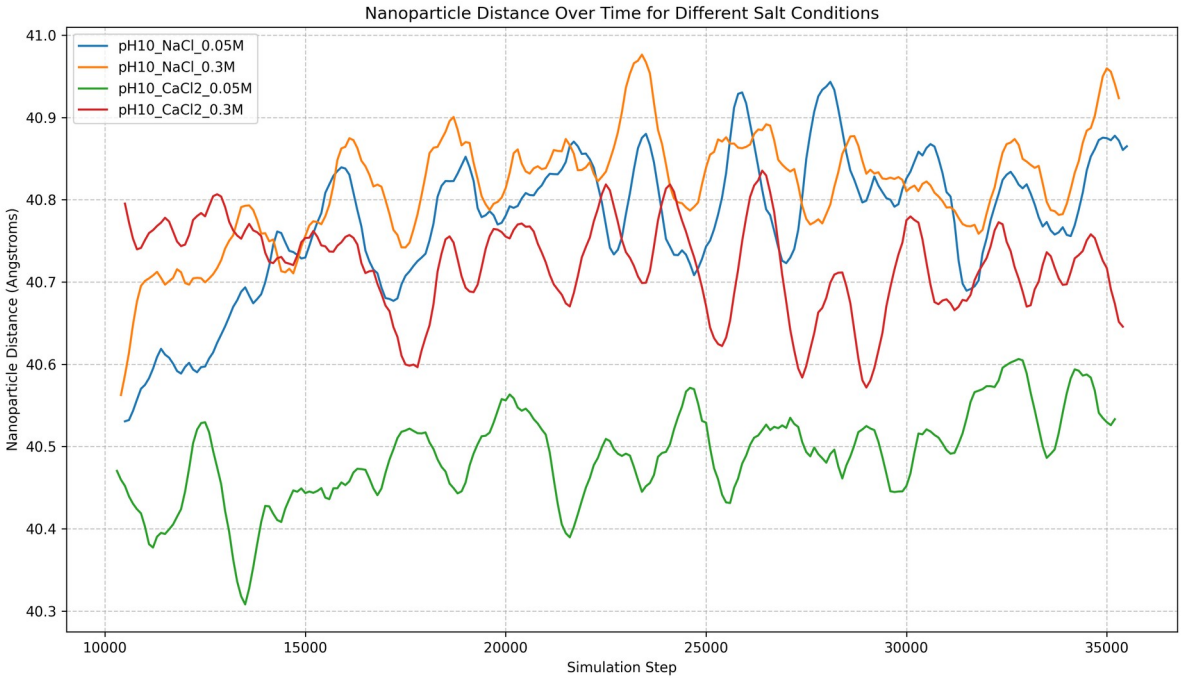



*Figure 8. Particle separation for the four salt conditions at pH 10. All traces stay within about 0.7 Å and cross each other.*

All four traces lie within 40.3–41.0 Å and overlap. $CaCl_2$ at 0.05 M sits lowest, but $CaCl_2$ at 0.3 M does not, so there is no consistent dependence on concentration or valence. This is unsurprising: the 468 counterions already give an effective concentration of roughly 6 M in the free volume, so 0.05–0.3 M of added salt barely changes the screening in this model. The salt study does **not** show a salt effect. A meaningful test needs a larger box, a realistic surface charge density, longer runs and replicas.

## 7. Limitations

- **Surface charge density is too high.** 234 charges over about 28 $nm^2$ is roughly 8 $e/nm^2$. Real silica at high pH typically carries about 1–2 $e/nm^2$ or less. The charge is a model of a fully ionised surface.
- **Small periodic box.** The box is only 42 Å across in y and z, so each particle is about 12 Å from its own periodic image. Image interactions along x partly cancel the net force between the two particles and lower the observed drift.
- **Short single runs.** Each condition is one trajectory of 35–55 ps. Drifts of 0.1–0.4 Å are comparable to the thermal fluctuation of 0.2–0.4 Å, and no uncertainty can be given from one run.
- **pH is a lookup, not a calculation.** The charge comes from a fixed pKa with random site selection. It ignores site heterogeneity and the dependence of pKa on local charge, and it saturates, so pH 10 and 12 (and pH 2 and 4) are the same system.
- **Real silica does not aggregate at low pH in water.** Near its isoelectric point (pH 2–3) silica sols are stable because of hydration forces. The model has no explicit mechanism for that, so the neutral runs show no net force, not an aggregation event.
- **Salt study is underpowered.** 3 to 21 ion pairs, borrowed ion parameters ($Cl^-$, $Ca^{2+}$), and a counterion background of several molar.
- **Timescale.** Aggregation of nanoparticles occurs over nanoseconds to seconds. The run lengths here probe only the early force between the particles.

## 8. Issues encountered and how they were solved

| Problem | Fix |
|---|---|
| $Na^+$ read as nitrogen by ASE | Rewrote the ion PDB in strict fixed-column format (element in columns 77–78) and added an atom-count assertion |
| "Must define bond_style before Bond Coeffs" | Declared pair, bond, angle and kspace styles before `read_data` |
| Over-coordinated O/Si after capping | Iterative outward nudging, with a Si–O-only coordination check |
| Energies of $-10^{16}$ and "Bond atoms missing" in salt runs | The SHAKE constraint line had been left out of the input; restored it and added a check that the line is present |
| Very slow single-thread runs (0.3 ns/day) | OpenMP with 2 threads, a smaller box, shorter runs |
| LAMMPS not found, Python/ASE mismatch in Colab | Reset the runtime, reinstall with condacolab, use `%pip` |

## 9. Conclusions and next steps

- A reproducible pipeline for building charged silica nanoparticles, setting pH-dependent charge and running two-particle MD in LAMMPS was assembled and used for four distinct charge states and four salt conditions.
- Neutral particles show no net drift; partly and highly charged particles drift apart by a few tenths of an Ångström, consistent with charge-induced repulsion. The pH 7 result is the clearest.
- At full charge the extra drift is not larger, and $Na^+$ binding is stronger; condensation is a plausible but untested explanation.
- The salt runs do not resolve any effect, for reasons listed above.

**To make the results quantitative:** (1) lower the surface charge to a realistic 1–2 e/nm²; (2) use a larger box; (3) compute the potential of mean force between the particles (umbrella sampling) instead of watching free drift; (4) run replicas and give error bars; (5) use a larger number of salt ions and ion parameters fitted for CLAYFF; (6) compare with zeta-potential data on silica or silica-sol gelling data for castables.

## 10. Files

The scripts, figures and README are in the accompanying folder. Run order: `01_build_nanoparticle.py` → `02_assign_pH_charges.py` → `03_build_system_and_solvate.py` → `04_generate_lammps_inputs.py` → LAMMPS → `07_plot_results.py`, `05_na_rdf_analysis.py`; salt study with `06_salt_concentration_study.py`; visualisation with `08_visualize_ovito.py`.